

# Alluvial Bed Forms and Flow Resistance

## Mechanisms, Classifications, and Engineering Applications

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### Executive Summary

Alluvial bed forms—such as ripples, dunes, and antidunes—are morphodynamic structures that strongly influence hydraulic resistance and sediment transport in sand-bed rivers. These bed forms arise from self-organized instabilities produced by sediment–flow interactions.

A critical engineering principle in alluvial channels is that **total bed shear stress**  $\tau_b$  is not the effective force governing sediment transport. Instead, total resistance must be partitioned into:

$$\tau_b = \tau'_b + \tau''_b$$

where:

- $\tau'_b$  = grain shear stress (effective for sediment motion)
- $\tau''_b$  = form drag (caused by pressure differences across bed forms)

Only the grain component  $\tau'_b$  contributes to sediment transport.

This briefing summarizes:

- Bed-form classification across hydraulic regimes
- The physical instability mechanism governing bed-form evolution (Exner concept)
- Three major resistance-partitioning approaches:
  - Einstein–Barbarossa (depth decomposition)
  - Engelund (slope decomposition)
  - Modern algebraic friction-factor implementation

Accurate regime identification and consistent resistance partitioning are essential for reliable prediction of rating curves and sediment transport rates. In dune-dominated flows, form drag may dominate total resistance, substantially increasing stage for a given discharge.

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# 1. Classification and Characteristics of Bed Forms

Bed forms are classified according to the hydraulic regime in which they develop. The governing parameter is the **Froude number**:

$$Fr = \frac{U}{\sqrt{gh}}$$

which distinguishes gravity-wave-dominated flow from inertia-dominated flow.

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## Bed Form Classification by Regime

| Flow Regime | Bed Form         | Phase Relation (Bed vs Surface) | Typical $Fr$   | Primary Resistance     |
|-------------|------------------|---------------------------------|----------------|------------------------|
| Lower       | Ripples          | Out of phase                    | $Fr < 1$       | Form roughness         |
| Lower       | Dunes            | Out of phase                    | $Fr < 1$       | Form drag (separation) |
| Transition  | Washed-out dunes | Variable                        | $Fr \approx 1$ | Variable               |
| Upper       | Plane bed        | In phase                        | $Fr \approx 1$ | Grain roughness        |
| Upper       | Antidunes        | In phase                        | $Fr > 1$       | Wave dynamics          |
| Upper       | Chutes/Pools     | In phase                        | $Fr > 1$       | Continuous transport   |

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## Characteristic Profiles

### Ripples

- Small-scale: wavelength and height  $\ll h$
- Common in hydraulically smooth or transitional flow
- Downstream migration
- Dominated by grain-scale processes

## Dunes

- Large-scale: height  $\approx 0.1\text{--}0.3h$
- Wavelength  $L_d \approx 6\text{--}7h$
- Asymmetrical shape with slip face
- Strong flow separation and pressure drag

## Antidunes

- Occur in supercritical flow
- Strongly influenced by free-surface wave dynamics
- May migrate upstream or downstream

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# 2. Morphodynamic Instability and the Exner Concept

Bed forms arise from instabilities in the coupled sediment–flow system. Their evolution is governed by the **Exner equation**:

$$\frac{\partial \eta}{\partial t} + \frac{1}{1 - \lambda_p} \frac{\partial q_b}{\partial x} = 0$$

where:

- $\eta$  = bed elevation
- $q_b$  = volumetric sediment transport
- $\lambda_p$  = bed porosity
- $x$  = streamwise coordinate

## Instability Mechanism

1. A small bed perturbation creates a shear-stress perturbation.
2. Shear perturbation causes a transport perturbation.
3. A phase lag between bed topography and transport leads to growth of the bed form.

This feedback loop produces self-organized patterns such as dunes and ripples.

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### 3. Flow Regime Identification Tools

Correct resistance closure depends on identifying the hydraulic regime.

#### Key Diagnostic Parameters

##### Froude Number

$$Fr = \frac{U}{\sqrt{gh}}$$

Distinguishes subcritical (dune) from supercritical (antidune) flow.

##### Shields Parameter

$$\Theta = \frac{\tau_b}{(\rho_s - \rho)gD}$$

Indicates sediment mobility and transport stage.

##### Relative Submergence

$$\frac{h}{d}$$

Represents separation between flow depth and roughness scale.

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### 4. Theory of Resistance Partitioning

In alluvial channels:

$$\begin{aligned}\tau_b &= \tau'_b + \tau''_b \\ f &= f' + f''\end{aligned}$$

#### Critical Insight

- $\tau'_b$  (grain shear) drives sediment transport.
- $\tau''_b$  (form drag) results from pressure differences across bed forms.

- Form drag does **not** initiate particle motion.

Using total shear  $\tau_b$  in sediment-transport formulas overpredicts transport in dune regimes.

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## 5. Comparison of Resistance Partitioning Methods

Three major approaches are used in engineering practice.

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### Method Overview

| Method                     | Basis of Decomposition          | Empirical Closure    | Implementation        |
|----------------------------|---------------------------------|----------------------|-----------------------|
| Einstein–Barbarossa (1952) | Depth: $R_b = R'_b + R''_b$     | Fig. 6.27            | Graphical / Iterative |
| Engelund (1966)            | Slope: $S = S' + S''$           | Fig. 6.28            | Graphical / Iterative |
| Modern Algebraic           | Friction factor: $f = f' + f''$ | Roughness predictors | Numerical             |

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### Einstein–Barbarossa (1952)

**Core idea:** split hydraulic radius, not slope.

- Grain roughness:  $k'_s = D_{65}$
- Log-law with transitional correction factor  $\chi$
- Form component from empirical  $U/u''_* - \Psi$  relation
- Often predicts strong form dominance in dune regime

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### Engelund (1966)

**Core idea:** split slope into grain and form components.

- Grain roughness:  $k'_s = 2D_{50}$
- Empirical relation between total and grain Shields:

$$\theta' = 0.06 + 0.4\theta^2 (\text{lower regime})$$


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## Modern Algebraic Friction-Factor Partition

- Uses Darcy–Weisbach framework
- Grain friction based on  $k'_s \approx 2-3D_{90}$
- Total roughness:

$$k_s = k'_s + k_{s,\text{form}}$$

- Dune geometry closure:

$$k_{s,\text{form}} = 11\Delta_d(1 - e^{-25\Delta_d/L_d})$$

Commonly implemented in:

- HEC-RAS
  - Delft3D
  - TELEMAC
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## 6. Engineering Applications and Observed Trends

### Rating Curve Behavior

In sand-bed rivers:

- Dunes increase  $f'' \rightarrow$  higher stage for given discharge.
- As dunes wash out (transition to plane bed),  $f''$  drops rapidly.
- Stage may decrease even as discharge increases.

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## Comparative Example

( $Q = 60 \text{ m}^3/\text{s}$ ,  $S = 0.0004$ )

| Metric                | Einstein–Barbarossa | Engelund | Modern Algebraic |
|-----------------------|---------------------|----------|------------------|
| Depth $h$             | 2.558 m             | 1.980 m  | 2.362 m          |
| Grain Shear $\tau'_b$ | 1.38 Pa             | 2.46 Pa  | 1.48 Pa          |
| Form Shear $\tau''_b$ | 8.65 Pa             | 5.31 Pa  | 7.79 Pa          |
| Grain Fraction        | 0.14                | 0.32     | 0.16             |

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## Key Insights

### Depth Prediction

$$h_{\text{Engelund}} < h_{\text{Modern}} < h_{\text{E\&B}}$$

### Grain Fraction Variation

Differences by a factor of two are common. Because:

$$q_b \propto (\tau'_b - \tau_c)^n$$

small differences in  $\tau'_b$  can lead to large transport discrepancies.

### Regime Consistency

All methods correctly identify the lower-regime dune configuration.

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## 7. Essential Modeling Takeaways

1. Form drag dominates in lower-regime flows.
2. Use grain shear for sediment transport calculations.
3. Partitioning is less critical in upper-regime plane-bed flows.
4. Hydraulics and resistance partition must be iterated consistently.